



Methodology to assess performance indexes for sensitivity, selectivity, and speed of coordination of directional overcurrent protections

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ABSTRACT

Many algorithms for the coordination of directional overcurrent protections (DOCP) have been proposed in literature claiming that optimal solutions are reached, even though there are different reported solutions for the same problem. A new method to evaluate the quality of solutions for the coordination of DOCP is proposed. Three indexes are defined for selectivity, sensitivity, and speed, as desirable features of the coordination of protections. The use of performance indexes to evaluate the response of DOCP systems has not been previously considered in the literature. Given time dial and pick-up settings provided by any program or algorithm, uniformly distributed fault locations are simulated in each transmission line in order to determine the operation times as well as the separation times between main and backup relays. For each fault, the trip sequence of each relay pair is identified and classified according to 15 possible outcomes. To illustrate the method, indexes were computed from seven different coordination solutions reported in literature over the same test system. Results show interesting trade-offs among the three indexes, suitable to be presented as three independent axes; consequently, a given solution could be better in one feature (e.g., speed) but worse in others (e.g., selectivity and/or sensitivity).

1. Introduction

The use of directional overcurrent protections is widespread in electric power systems industry. Economy and simplicity justify the use of the overcurrent function. Nevertheless, the coordination of directional overcurrent protections (DOCP) is still a challenging task for engineers.

The coordination of DOCP in interconnected systems must be regarded as multiple criteria problem where several and different desired characteristics as sensitivity (fault detection), selectivity (minimizing zones out-of-service) and speed (fast fault clearance) must be simultaneously considered in order to find the “best” problem solution [1]. In fact, there is no single solution, but several possible solutions depending on the objective to be optimized.

We can find in literature a huge number of optimization methods to coordinate DOCP. Comprehensive reviews on this topic can be found in [2–6]. Existing contributions solve the optimization problem using a single objective formulation, usually by maximizing primary relay speeds (or minimizing relay operation times). In general, these optimization problems have been formulated considering only few fault locations (e.g., near-end, far-end, and mid-point) and disregarding transient network configurations during the relays’ tripping sequences. The main differences between existing proposals lie on the method used

to solve the optimization problem. Some approaches use linear and non-linear programming [7]. Other approaches are based in evolutive techniques and metaheuristics [8].

Previous works on optimal coordination of DOCP did not validate their results for a wide range of fault locations and did not show the trade-offs among selectivity, speed, and sensitivity. The solutions provided by existing algorithms are strongly dependent on the set of fault locations chosen to formulate the optimization problem constraints [9]. Therefore, the proper solution should consider all the possible fault locations since the short-circuit current magnitudes and angles change along a transmission line. As a result, from existing literature in this topic, it is not clear the extent of how a given method for DOCP coordination fulfills sensitivity, selectivity, and speed objectives from a wide perspective. Simulation and verification are necessary to assess the appropriateness of relay coordination solutions [10]. Many fault locations should be considered in the simulation, and the results with one specific set of relay adjustments can be: (a) selective in some cases but not in others; (b) properly sensitive in some cases but not in others; (c) faster/slower, depending on fault location. Thus, the coordination should be evaluated from a statistical standpoint.

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In our view, existing optimal coordination of DOCP procedures are based on partial objective functions that cannot capture the expected behavior of the entire protection system. To illustrate this point, notice that many algorithms for coordination of DOCP have been proposed in literature claiming that optimal solutions are reached, even though there are different reported solutions for the same problem [11–16]. As these results are clearly dissimilar and will produce different outcomes when implemented in real-world, it is worth raising the following research question: Are existing optimization procedures for the coordination of DOCP capable to provide “optimal or efficient solutions” from the multiobjective optimization perspective? Can we have an evaluation tool to verify the quality of provided solutions?

So, to answer the raised research question, we think that DOCP settings obtained from any optimization procedure should be evaluated considering global performance indexes for a wide range of possible faults in the system. That is, each proposed optimal solution for this problem will produce specific results for sensitivity, selectivity, and speed, considering a wide range of possible faults. Proposed solutions can be considered mathematically adequate or not, depending on achievement of conditions like Pareto optimality, where a given solution cannot be changed to improve an objective (e.g., speed) without making worse at least another objective (e.g., selectivity or sensitivity) [17]. This multiobjective evaluation is essential to get rid of inefficient solutions.

In our knowledge the use of performance indexes to evaluate the response of DOCP systems has not been previously considered in the literature. This paper fulfills the research gap by introducing a new and general method to compute expected values that characterize selectivity, sensitivity, and speed of a complete DOCP system. The methodology allows us to benchmark and compare different solutions obtained by protection engineers for the same DOCP system. Thus, given time dial settings and pick-up settings provided by any program, algorithm or expert, uniformly distributed fault locations are simulated in each transmission line in order to determine the operation times of main relays as well as the separation times between backup and main relays. For each fault, the trip sequence of each relay pair is identified and classified according to 15 possible outcomes. The novelty of this proposal arises from the fact that we are capable to determine performance indexes for sensitivity, selectivity, and speed for any solution specified by existing or future coordination methods.

To illustrate the method, the proposed performance indexes were computed from seven different coordination solutions reported in literature [11–16] over the same test system [7]. The results show the existence of trade-offs among the proposed indexes, which are suitable to be presented as three independent axes and, consequently, a given solution could be better in one index but worse in other one. In addition, the proposed method allows the identification of inefficient solutions. This article also poses a new research challenge for the future: how to develop new optimization procedures for the coordination of DOCP using simultaneously the three mentioned objectives (selectivity, sensitivity, speed) and a wide range of possible faults in the analyzed system.

The rest of the paper is organized as follows. Section 2 discussed the background of the problem. Section 3 describes the methodology. Results are analyzed in Section 4. Conclusions are drawn in Section 5.

2. Background and related work

There are many contributions on optimal coordination of DOCP. In [4,5] we can find a list of around 80 relevant papers from 1988 to 2014. Recently, in [3,6], an updated survey of new papers published on the topic between 2014 and 2021 is presented. Existing contributions on optimal coordination of DOCP are based upon a single objective (usually relay speed) subject to selectivity and sensitivity constraints. The problem can be formulated as linear and non-linear depending on the nature of unknown variables. Solutions have been obtained

Table 1

Summary of existing validation procedures for DOCP coordination solutions.

Category	References
1. No validation	110 papers in [3–6]
2. Verification using simulation tools	[18–24]
3. Validation using performance indexes	There is no precedent

using a wide variety of optimization methods (e.g., linear, nonlinear, evolutionary).

In Table 1 we include the afore-mentioned reviews [3–6] as well as the papers that include verification with any kind of simulation tool [18–24]. Existing contributions are classified into three categories.

Category 1 lists contributions whose results are presented without any kind of verification using external tools/software. In this category there is no validation procedures of the results from selectivity and sensitivity goals as desired features of the protection system.

Category 2 shows existing contributions with verification carried out using specialized software and/or hardware under specific conditions. In this category, no validation procedures to evaluate the quality of the results are included.

Category 3 must comprise contributions that validate the results using a comprehensive statistical method based on global performance indicators, including mean and variances for metrics such as speed, sensitivity, and selectivity.

Vast majority of existing contributions on optimal coordination of DOCP are included in the first category with no procedures for verification or validation of their results. Regarding the second category, some contributions have verified their optimization results using a simulation software such as Digsilent [18] or ETAP [19]. In [20] verification of specific coordination violations is carried out using an analytical approach. Some approaches have applied real-time domain simulation tools to verify reported results: RTDS [21], OPAL-RT [22] and Matlab-Simulink [23] as well as other Hardware-in-the-loop schemes [24], mainly in meshed distribution systems. However, no methods have been reported to validate and characterize the performance of the protection system using statistics.

Regarding the third category, there is no precedent in the literature of an extensive validation of the selectivity using a wide range of fault locations, with assessment of the fact of having or not sensitivity for each fault, and the calculation of performance indexes to quantify the selectivity, sensitivity, and speed of the proposed solutions. This paper aims to fulfill this research gap. This topic is very important in case of modern power systems since the need of adaptive protection, based on optimal coordination of DOCP, has been well-established [24].

3. Methodology

In this section we present the proposed methodology. First, we indicate the main assumptions used to simulate faults and evaluate the response of the DOCP system for a given coordination solution. After the definition of the protection system model, we describe the methodology for the identification, analysis, and classification of relay pairs, as well as the structure of proposed performance indexes for selectivity, sensitivity, and speed.

3.1. Assumptions

The following assumptions were considered:

1. Only the 67 phase function is considered. Relay time dial and pickup settings as well as curve characteristics are known parameters.
2. Three-phase short-circuit current is determined using the Zbus approach [25]. Short circuit current magnitude is considered invariant over time and does not include load current contributions.

3. No pre-fault conditions are considered.
4. Conventional current transformer saturation is not considered.
5. Circuit breaker operation times are not included in the model.
6. Relay time operation considering transient network configurations is estimated according to [16,26].

Assumptions 1–5 are often considered in many references about optimization methods to coordinate DOCP. The consideration of transient configurations (assumption 6) to compute the operation times has been only adopted by very few previous papers [16,23,26], and it is herein adopted because is a more proper way to compute the operation times and is necessary for the proper classification of relays' behavior.

The assumptions listed above have been often applied in optimal coordination of DOCP and are oriented to determine the proposed indexes in a fair way, by considering the same premises that solutions to be herein compared as an example. A summary of results considering pre-fault conditions for the fault current calculations (i.e., without considering the assumption 3) is also herein included because the possible application of assumption 3 is not clearly described in some cases of published optimal solutions. As shown in Section 4.3, the influence of assumption 3 on the analyzed results is not significant, but it has been herein analyzed for the sake of rigorousness.

As the procedure to get performance indexes is general, the proposed methodology is also valid using circuit breakers operation times, load currents, current transformers saturation and the waveforms of instantaneous short-circuit currents obtained from any electromagnetic transient program (off-line or in real-time). The inclusion of more realistic conditions should improve the accuracy of classification procedures, yielding more realistic performance indexes, despite the need of applying detailed dynamic models for overcurrent functions and the difficulties to include all the relay details. These matters are out of scope of the paper. It is important to highlight that the proposed methodology can be applied to large-scale power grids since the method does not depend on the system size; that is, the fault currents can be computed using any commercial power system simulation software, capable to manage thousands of nodes and lines, and the assessment of proposed indexes can be easily performed with those input data.

3.2. Power system and protection system modeling

The interconnected power system is characterized by a set of transmission links (**K**), a set of transmission buses (**B**) and a set of generation buses (**G**). The total number of transmission lines, transmission buses and generation buses are given by $n_k = \dim(\mathbf{K})$, $n_b = \dim(\mathbf{B})$ and $n_g = \dim(\mathbf{G})$, respectively. Each generator has an equivalent impedance $z_g \in \mathbf{Z}_G$ that may include the step-up transformer impedance.

Fig. 1 displays a typical transmission line $k \in \mathbf{K}$ that connects a bus $b \in \mathbf{B}$ with another not specified transmission bus. Each link is represented by the π -model (a series impedance $z_k \in \mathbf{Z}_K$ and a shunt admittance $y_k \in \mathbf{Y}_K$).

Let us define a set **H** of three-phase faults at line k . The number of total fault locations are $n_f = \dim(\mathbf{H})$. Each element $h \in \mathbf{H}$ give us information about the location of the fault; h is the per unit line length from the left-hand circuit breaker associated with relay r shown in Fig. 1. For example, $h = 0.5 = 50\%$ means that the fault is located at midpoint of the line.

Each transmission line k is protected by two directional overcurrent relays (67P) located at line ends and polarized to detect faults within the line. In each line these two relays are denominated primary or main relays since they must detect the fault in the first instance. Furthermore, primary relays located at remote ends of adjacent lines, seen towards the fault, should be regarded as secondary or backup relays that must operate later if one or both primary relays do not operate for that fault or corresponding circuit breakers do not clear the fault current contributions.

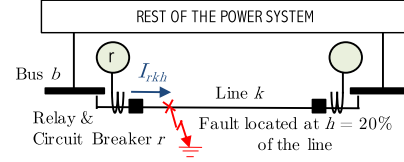


Fig. 1. Faulted transmission line k protected by a relay r .

Thus, the interconnected system has $r = 1, \dots, n_r = 2n_k$ relays suitable to be coordinated. The set of relays is defined as $r \in \mathbf{R}$ with $n_r = \dim(\mathbf{R})$. Relay r is characterized by the following manner. Time-current curve parameters are known for each relay. There are many types of possible inverse curves. For the sake of simplicity, only standardized IEC curves are considered in the numerical examples, which are defined by two parameters: $[\alpha_{r1}, \alpha_{r2}] \in \alpha_R$, and all the relays have the same curve type. However, any kind of curve can be simulated for any relay location with the proposed methodology. Each relay r has two settings: (a) the time dial setting $D_r \in \mathbf{D}$; (b) the pickup current setting $P_r \in \mathbf{P}$. Both settings are traditionally selected by protection engineers using coordination criteria, and they can be selected with the help of an optimization tool.

In summary, the data sets required for the method are generators (**G**), buses (**B**), line links (**K**), line models ($\mathbf{Z}_K$ and $\mathbf{Y}_K$), generator models ($\mathbf{Z}_G$), and the set of faults locations (**H**) for power system modeling, and relay parameters and settings (**R**, α_R , **D**, and **P**), for protection system modeling.

The operation time of each relay is computed according to the following expression, in case of standardized IEC curves:

$$T_{rkh} = \frac{\alpha_{r1} D_r}{\left(\frac{I_{rkh}}{P_r}\right)^{\alpha_{r2}} - 1} = \beta_{rkh} D_r \quad (1)$$

where $T_{rkh} \in \mathbf{T}$ is the operation time of relay r in seconds, and $I_{rkh} \in \mathbf{I}$ is the rms current flowing through the relay r for a fault at location h of line k . As indicated in Section 3.1, only three-phase solid faults are herein considered for the sake of simplicity, but the proposed method can be easily extended to faults through resistance.

Fault current magnitudes and angles can be obtained using approximated methods or advanced multi-phase simulators. In this case, we use the Zbus approach [25] to determine short-circuit currents flowing through each relay r of the system.

Parameter $\beta_{rkh} \in \beta$ is a linear coefficient that multiplies the time-dial setting D_r in order to obtain the operation time of relay r . This linear parameter is crucial to perform the relay-pair classification procedure. Notice that a negative β_{rkh} implies a loss of sensitivity because $I_{rkh} < P_r$. The angle $\theta_{rkh} \in \Theta$ associated with the current flowing through relay r is required to verify whether the direction of the current is in the operation zone for the directional element, which is an important verification in case of backup relays.

Notice in Fig. 1 that for a fault located at h , the series impedance and shunt admittances of the faulted line are split in two parts: $z_{left} = h \cdot z_k$, $y_{left} = h \cdot y_k$ and, $z_{right} = (1 - h) \cdot z_k$, $y_{right} = (1 - h) \cdot y_k$, respectively. Fault location h is then included as an additional bus into the augmented system admittance matrix ($\mathbf{Y}_{bus}$) required to compute short circuit contribution.

3.3. Relay pair identification and analysis

In interconnected power systems each primary/main relay of a transmission line must have one backup relay in each adjacent line. All the system main-backup relay pairs must be identified of the system for each possible case of fault in transmission lines. For instance, if we

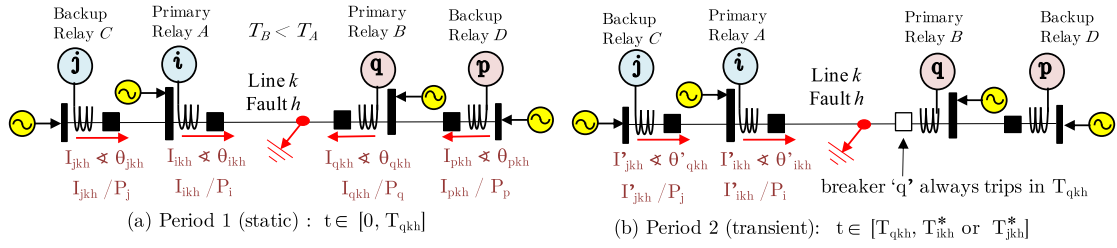


Fig. 2. Relay pair analysis for a fault at location h in line k .

consider a system with four transmission buses ($n_b = 4$) and all the possible single-line connections among them (six single transmission lines; $n_k = 6$), there are four backup-main relay pairs for each possible fault in a transmission line. Thus, a set of pair relays can be identified as $\mathbf{NP} = [p(1), \dots, p(k), \dots, p(n_k = 6)] = [4, 4, 4, 4, 4, 4]$. The total number of backup-main relay pairs is $n_{pr} = 6 \cdot 4 = 24$ for this example. This basic system has the combination of $n_r = 2n_k = 12$ relays operating as primary with 24 relays as backup.

Fig. 2(a) shows a transmission line k with two primary relays A and B. Primary relay A has a backup relay C located at the end of an adjacent line. Primary relay B has a backup relay D located at the end of an adjacent line.

When a fault is simulated at location h of line k , there are two possibilities:

1. Relay B operates faster than A ($T_B < T_A$) and the circuit breaker B effectively clears the fault current contribution, then the tag “q” is assigned to relay B ($q \rightarrow B$), tag “i” to relay A ($i \rightarrow A$), tag “p” to relay D ($p \rightarrow D$) and tag “j” to relay C ($j \rightarrow C$). This is just the case depicted in Fig. 2.
2. Relay A operates faster than B ($T_A < T_B$) and the circuit breaker A effectively clears the fault current contribution, then the tag “q” is assigned to relay A ($q \rightarrow A$) and so on: $i \rightarrow B$, $p \rightarrow C$ and $j \rightarrow D$. This case is not shown in Fig. 2, for the sake of shortness.

The analysis determines, in all lines k of the system and for every fault location h , who is the primary relay tagged as “q” (the faster main relay) and who is “i” (the slower main relay). Accordingly, relay “p” is regarded as the backup of “q”, and relay “j” is regarded as the backup of “i”. Once relays q, i, p , and j are duly tagged for each fault location h at line k , the corresponding operating times should be computed according to Eq. (1).

Relay q is the faster main relay, in Fig. 2. After the circuit breaker opening, at relay q location, the system topology changes as shown in Fig. 2(b), and there is a fault current contribution still active from relays i and j . As a result, a transient network topology appears until the opening of a circuit breaker associated with primary relay i or backup relay j . During the transient condition, a new system model must be determined in order to recalculate current contributions (magnitudes and angles). In the remainder of the paper, transient variables are denoted with apostrophe (').

Thus, the fault clearing process comprises two periods: the static one and the transient one. Hereafter we refer these two periods as period 1 and period 2. The period 1 lasts until breaker q clears the fault ($0 \leq t \leq T_{qkh}$). This period corresponds to a static condition represented by an admittance matrix $\mathbf{Y}_{bus}$ where all lines segments are connected. Once the breaker q clears the fault, a transient condition arises until the operation of breaker i or breaker j (if i is not faster than j). Thus, period 2 lasts until breaker i ($T_{qkh} \leq t \leq T_{ikh}^*$) or breaker j ($T_{qkh} \leq t \leq T_{jkh}^*$) clears the correspondent fault current contribution. During period 2, the system should be represented by a new admittance matrix $\mathbf{Y}'_{bus}$ that models the transient network produced by the line trip-out associated to breaker located at q . As relays i and j are affected by currents of period 1 and period 2, the effective operating times T_{ikh}^* and T_{jkh}^* should be determined using the formulation given in [26].

In each period, short-circuit studies must be carried out to determine fault current contributions. Notice that each fault location h at line k and relay r is represented in magnitude (I_{rkh} and I'_{rkh}) and angle (θ_{rkh} and θ'_{rkh}) under static (period 1) and transient (period 2) conditions. As the relay settings are known, Eq. (1) can be applied to determine the linear coefficients (β_{rkh} and β'_{rkh}) and the operation times under static and transient conditions (T_{rkh} and T'_{rkh}) for $r = q, i, p$, and j .

In this paper we assume that at least one main relay (relay q) is faster than the corresponding backup relays. For the sake of simplicity, we do not consider the unlikely case of having the two primary relays of a line slower than the correspondent backup relays. This theoretical possibility corresponds to an extremely bad coordination, which does not occur in none of the herein analyzed cases.

On the other hand, it is herein assumed that the breaker of the faster primary relay q always clears the fault contribution. Conversely, if the faster primary relay q is sensitive but its breaker does not clear the fault (i.e., breaker failure condition), the next transient network configurations will depend on operation time of relays i, j or p , and the operation times of relays i and j during period 2 are not as described ones. For the sake of simplicity, the proposed methodology does not consider the described breaker failure condition.

It is necessary to mention that a relay i (the slower main relay) can have two or more back-up relays and one of them may be faster than relay i . This is an undesired condition but it is possible. In this case, the comparison of operation times between the relay i and the remaining back-up relay(s) is influenced by another transient configuration, which is not herein considered for the sake of simplicity (the proposed assessment of the relay pairs is good enough without considering this complexity). That is, the comparison between operation times of relays i and j is herein performed considering only the transient configurations depicted in Figs. 2(a) and 2(b).

3.4. Relay pair classification

In Fig. 2(a) we analyze two kind of relay pairs: $i-j$ and $q-p$. When a fault is simulated at location h of line k we can classify the behavior of each relay pair according to two classes, 6 types and 15 sub-types, as shown in Fig. 3 and summarized in the following list:

1. Class F – Feasible calculation of a separation time

- (a) Type 1: Both relays of pair $q-p$ advance towards the trip condition during period 1 (case 1).
- (b) Type 2: Both relays of pair $i-j$ advance towards the trip condition during period 1 and period 2 (case 2)
- (c) Type 3: The DOCP at relay j does not advance towards the trip condition during period 1
 - i. Sub-Type 3a: Relay j does not advance only due to no sensitivity (case 3)
 - ii. Sub-Type 3b: Relay j does not advance only due to current in the opposite direction (case 4)
 - iii. Sub-Type 3c: Relay j does not advance due to current in the opposite direction and no sensitivity (case 5)

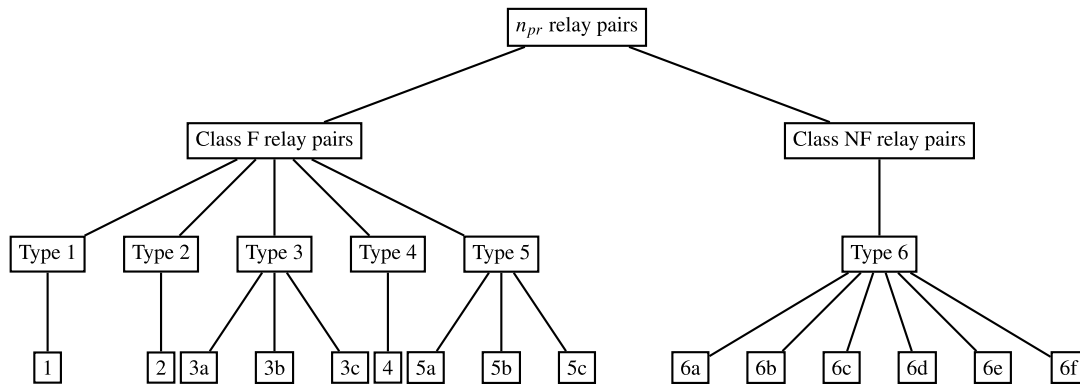


Fig. 3. Classification of relays pairs according to the calculation of S .

(d) Type 4: The DOCP at relay i does not advance towards the trip condition during period 1 due to no sensitivity (case 6)

(e) Type 5: The DOCP at relays i and j do not advance towards the trip condition during period 1:

- i. Sub-Type 5a: Relays i and j do not advance only due to no sensitivity (case 7)
- ii. Sub-Type 5b: Relay i does not advance due to no sensitivity and relay j does not advance due to current in the opposite direction (case 8)
- iii. Sub-Type 5c: Relays i and j do not advance due to no sensitivity and, additionally, relay j does not advance due to current in the opposite direction (case 9)

2. Class NF – No feasible calculation of a separation time

(a) Type 6: A backup relay cannot perform the backup function

- i. Sub-Type 6a: Relay p does not advance only due to current in the opposite direction in period 1 (case 10)
- ii. Sub-Type 6b: Relay j does not advance only due to current in the opposite direction in period 2 (case 11)
- iii. Sub-Type 6c: Relay p does not advance only due to no sensitivity in period 1 (case 12)
- iv. Sub-Type 6d: Relay p does not advance due to current in the opposite direction and no sensitivity in period 1 (case 13)
- v. Sub-Type 6e: Relay j does not advance due to no sensitivity and current in the opposite direction in period 2 (case 14)
- vi. Sub-Type 6f: Relay j does not advance only due to no sensitivity in period 2 (case 15)

A relay pair is classified as “F” if a *separation time* is suitable to be calculated. Otherwise, the relay pair is classified as “NF”. The separation time $S \in \mathbf{S}$ is defined as the lapse between backup and primary relay operation times: $S = T_{\text{backup}} - T_{\text{main}}$ where T_{main} is the operation time of the primary relay of line k for a fault located at location h of line k , T_{backup} is the operating time of a backup relay located in an adjacent line. In the remainder of the paper, we consider T_{jkh} and T_{pkh} as operation times of backup relays, and T_{ikh} and T_{qkh} as operation times of primary relays, and both times stored as elements of set $\mathbf{T}$.

The sub-classification of F-pairs and NF-pairs are based on the reasons that can avoid the advance of a DOCP: (a) no sensitivity; (b) current in the opposite direction. The no sensitivity condition occurs

when the current magnitude seen by the relay r is lower than the pick-up current ($I_{rkh} \leq P_{rkh}$ or $I'_{rkh} \leq P_{rkh}$) at period 1 or 2, respectively. This means that a negative linear factor β_{rkh} or β'_{rkh} is achieved at the time period of interest. On the other hand, the condition of current in the opposite direction happens when the corresponding current angle is out of the angular tripping zone. For instance, a reverse fault current condition can be identified in Fig. 2 if a current angle (e.g., θ_{rkh} or θ'_{rkh}) is out of the directional zone DZ (DZ is featured by a range of admissible angles for the directional detection of forward faults).

The relay sensitivity can be improved by reducing the pickup setting. This reduction in the pickup settings is bounded by the load currents, and in some cases fault current magnitudes are near to zero or are flowing in the opposite direction. Therefore, a complete sensitivity for all possible fault locations in the system is sometimes unattainable. However, the condition of current in the opposite direction is inherent to the direction of fault currents and cannot be improved by the selection of pickup settings. Due to this fact, the number of cases with loss of sensitivity only includes herein those cases of no sensitivity and without current in the opposite direction (i.e., cases whose sensitivity could be improved by changes in the pickup settings).

Within the two major classes defined above (F and NF relay pairs) we can characterize six relay pair types and 15 relay sub-types/cases as seen in Fig. 3. Table 2 indicate the conditions that must be accomplished in order to a given relay pair $i-j$ or $q-p$ be considered as Type 1 to 6 according to the taxonomy presented in Fig. 3.

Relay pair classification as F or NF must be carried out for $f = 1, \dots, n_f$ uniformly distributed faults per line. Thus, $\mathbf{N}$ is the set of analyzed relay pairs for the simulated faults, and the total number of analyzed relay pairs' behaviors is $N = \dim(\mathbf{N}) = n_{pr} n_f$ where n_{pr} is the number relay pairs of the system. Notice that N must equal the sum of analyzed faults sub-classified and accounted per sub-type ($N = \sum_{c=1}^{15} N_c$; N_c is identified at the last row of Table 2). Once the N analyzed relay pairs' behaviors are classified in classes, types and subtypes, three relevant subsets should be determined:

(1) The subset $\mathbf{S}$ where a relay-pair is classified as F, that is, analyzed faults with a calculated separation time between a given primary relay and its backup. The total number of analyzed relay pairs' behaviors (N_F) in this subset is

$$N_F = \dim(\mathbf{S}) = \sum_{c=1}^9 N_c. \quad (2)$$

On the other hand, the total number of analyzed relay pairs' behaviors (N_{NF}) where separation time cannot be calculated is $N_{NF} = N - N_F$.

(2) To evaluate selectivity, we must determine the subset $\boldsymbol{\varphi} \subset \mathbf{S}$ that accounts analyzed faults where relay pairs F have separation times below the prescribed safety margin $S < S_M$. The number of no selective relays is given by:

$$N_{\text{nosel}} = \dim(\boldsymbol{\varphi}) \quad (3)$$

Table 2
Relay pair classification.

Type	Class F relay pairs									Class NF relay pairs					
	1	2	3			4	5			6					
Sub-type			3a	3b	3c		5a	5b	5c	6a	6b	6c	6d	6e	6f
θ_{pkk} is out of the DZ?	No	–	–	–	–	–	–	–	–	Yes	–	No	Yes	–	–
θ_{jkh} is out of the DZ?	–	No	No	Yes	Yes	No	No	Yes	Yes	–	–	–	–	–	–
θ'_{jkh} is out of the DZ?	–	No	No	No	No	No	No	No	No	–	Yes	–	–	yes	no
β_{qkh} is negative?	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No
β_{pkk} is negative?	No	–	–	–	–	–	–	–	–	No	–	Yes	Yes	–	–
β_{ikh} is negative?	–	No	No	No	No	Yes	Yes	Yes	Yes	–	–	–	–	–	–
β'_{ikh} is negative?	–	No	No	No	No	No	No	No	No	–	–	–	–	–	–
β_{jkh} is negative?	–	No	Yes	No	Yes	No	Yes	No	Yes	–	–	–	–	–	–
β'_{jkh} is negative?	–	No	No	No	No	No	No	No	No	–	No	–	–	Yes	Yes
Cases per sub-type	N_1	N_2	N_3	N_4	N_5	N_6	N_7	N_8	N_9	N_{10}	N_{11}	N_{12}	N_{13}	N_{14}	N_{15}

(3) The subset of analyzed faults NL where backup relays cannot perform the backup function only due to no sensitivity in periods 1 or 2. The number of analyzed faults in this subset is herein named subset of analyzed faults with no sensitivity and it is given by:

$$N_{nosen} = \dim(\mathbf{NL}) = N_3 + N_7 + N_{12} + N_{15} \quad (4)$$

Eq. (4) accounts both partial and complete no sensitivity conditions. A more specific subset could contain only backup relays that cannot perform the backup function exclusively due to no sensitivity in period 2. In this case, $N_{nosen-option 2} = N_{15}$.

Notice that cases where backup relays cannot perform the backup function due to current in the opposite direction are not considered in subset NL. In such cases, the problem is inherent to the directionality of fault currents and cannot be solved by the selection of pickup currents.

3.5. Calculation of separation times and operation times

The 6 types of relay pairs, classified in the previous section, are featured by the way of computing the separation between the operation times. Next, we define the mathematical process to calculate operation and separation times for each type of relay pairs.

3.5.1. Type 1: Both relays of pair $q-p$ advance towards the trip condition during period 1 (case 1)

The separation time S_{pqk} is computed from the operation times T_{pkk} and T_{qkh} , which are defined by Eq. (1).

$$S_{pqk} = T_{pkk} - T_{qkh} \quad (5)$$

3.5.2. Type 2: Both relays of pair $i-j$ advance towards the trip condition during period 1 and period 2 (case 2)

The separation time S_{jikh} is the effective operation time T_{jkh}^* minus the effective operation time T_{ikh}^* . The effective operation times take into account the transition between the static mode depicted in Fig. 2(a) and the transient mode depicted in Fig. 2(b). According to [16,26]:

$$T_{jkh}^* = T_{qkh} + T'_{jkh} \left(1 - \frac{T_{qkh}}{T_{jkh}}\right) \quad (6)$$

$$T_{ikh}^* = T_{qkh} + T'_{ikh} \left(1 - \frac{T_{qkh}}{T_{ikh}}\right) \quad (7)$$

$$S_{jikh} = T_{jkh}^* - T_{ikh}^* = \beta'_{jkh} D_j - \beta'_{ikh} D_i - \gamma_{qjikh} D_q \quad (8)$$

$$\gamma_{qjikh} = \beta_{qkh} \left[\frac{\beta'_{jkh}}{\beta_{jkh}} - \frac{\beta'_{ikh}}{\beta_{ikh}} \right] \quad (9)$$

where T_{qkh} , T_{ikh} , T_{jkh} , T'_{jkh} , T'_{ikh} , β_{jkh} , β_{ikh} , β_{qkh} , β'_{ikh} , and β'_{jkh} are entries determined according Eq. (1).

3.5.3. Type 3: The DOCP at relay j does not advance towards the trip condition during period 1 (cases 3–5)

In type 3, relay j does not advance in period 1, then the separation time S_{jikh} should be calculated using the equations of Section 3.5.2, but considering that the quotient β'_{jkh}/β_{jkh} is zero in the calculation of γ_{qjikh} [16], and T_{jkh}^* is simply equal to T'_{jkh} .

3.5.4. Type 4: The DOCP at relay i does not advance towards the trip condition during period 1 (case 6)

In type 4, relay i does not advance in period 1, then the separation time S_{jikh} should be calculated using the equations of Section 3.5.2, but considering that the quotient β'_{ikh}/β_{ikh} is zero in the calculation of γ_{qjikh} [16], and T_{ikh}^* is simply equal to T'_{ikh} .

3.5.5. Type 5: The DOCP at relays i and j do not advance towards the trip condition during period 1 (cases 7–9)

In type 5, relays i and j do not advance in period 1, then the separation time S_{jikh} should be calculated using the equations of Section 3.5.2, but considering that γ_{qjikh} is zero, T_{jkh}^* is simply equal to T'_{jkh} , and T_{ikh}^* is simply equal to T'_{ikh} .

3.5.6. Type 6: A backup relay cannot perform the backup function (cases 10–15)

In type 6, the separation time for the relay-pair cannot be calculated.

3.6. Proposed performance indexes to assess the coordination of DOCP

In this section, we present the proposed performance indexes for selectivity, sensitivity, and speed of DOCP coordination.

3.6.1. Sensitivity index

The sensitivity index is defined as the portion of analyzed faults where relays can perform the backup function without sensitivity problems:

$$\text{sensitivity index} = 1 - \frac{N_{nosen}}{N} \quad (10)$$

where N is the total number of analyzed relay pairs' behaviors for n_{pr} backup relays across n_f fault locations ($N = n_{pr} n_f$). N_{nosen} accounts no-sensitive backup relays according to Eq. (4).

The meaning of proposed index is as follows: consider a sensitivity index of 0.9, This result implies that in 10% of N analyzed relay pairs' behaviors, the backup relays cannot detect the fault current contribution. This sensitivity index is dependent on pick-up settings. The maximum sensitivity index is obtained with the minimum pick-up settings, which are constrained by the load currents, as described in Section 3.4. That is, the protection engineer can improve the DOCP sensitivity by the selection of appropriate pick-up settings.

3.6.2. Selectivity index

The selectivity index is defined as the portion of analyzed faults for relay pairs class F whose separation time S is greater than a prescribed safety margin S_M .

$$\text{selectivity index} = 1 - \frac{N_{\text{nosel}}}{N_F} \quad (11)$$

The meaning of proposed index is as follows: consider a selectivity index of 0.9, This result implies no-selectivity in 10% of the N_F analyzed faults for relay pairs class F.

Two additional indexes related with selectivity will be reported, the average separation time and the minimum separation time. The average separation time is defined as $\bar{S} = \frac{1}{N_F} \sum_{s=1}^{N_F} S_s$. The minimum and maximum separation time are defined as $S_{\min} = \min(S)$ and $S_{\max} = \max(S)$, respectively. If this minimum is greater than the safety margin ($S_{\min} \geq S_M$) we can assure that the protection system has full selectivity (selectivity index of 100%) for all possible faults of the system.

3.6.3. Average operation time and speed index

The average operation time for $n_r=2n_k$ primary relays of the system and n_f faults is given by:

$$\bar{T} = \frac{1}{2n_k n_f} \sum_{h=1}^{n_f} \sum_{k=1}^{n_k} [T_{ikh}^* + T_{qkh}] \quad (12)$$

where relays i and q correspond to the relays located at each end of line k . Operation times $T_{qkh} \in \mathbf{T}$ and $T_{ikh}^* \in \mathbf{T}^*$ are calculated according to Section 3.5.

The average speed index is defined as follows:

$$\text{speed index} = \bar{v} = \frac{1}{2n_k n_f} \sum_{h=1}^{n_f} \sum_{k=1}^{n_k} \left[\frac{1}{T_{ikh}^*} + \frac{1}{T_{qkh}} \right] \quad (13)$$

3.7. Exhaustive-assessment algorithm to compute performance indexes of coordination of DOCP systems

The general procedure to calculate the proposed performance indexes of coordination of DOCP is presented in Fig. 4. Data setup is defined in row 1. Input data elements are grouped in three blocks: (a) power system model (sources, system impedances, and topology); (b) settings of relays (datatable with relay pair per line, relay curve types, and pickup and time dial settings); (c) coordination pairs (map of relay-pairs associated to each transmission line; each primary relay in line k has $p(k)$ backup relays that should be identified).

The algorithm has three loops. The first one assigns the counter f to analyze n_f fault locations h . The second one assigns the faulted line k . The third loop assigns the relay pair s of line k . Indexes f , k , and s are positive integer. Uniformly distributed fault locations h are assigned in per unit of total line length in row 3. In rows 5 to 25, all relays belonging to a relay pair s at faulted line k in location f are identified, analyzed, and classified. To do so, short-circuit contributions and relay operation times should be determined in order to identify the tripping sequence in Period 1. Each relay pair is then tagged as $i-j$ or $q-p$. In period 2, corrected operation times at primary and backup relays are calculated considering transient configurations. The resulting linear parameters β , β' , and γ , as well as current contribution angles θ are necessary to classify each relay pair s according to $c=1, \dots, 15$ cases (or subtypes) defined in Table 2.

Results obtained are aggregated after the end of each loop in rows 27–40. In line 41, the subset of relay pairs class F as well as the subsets of backup relays with no selectivity and no sensitivity are determined. Statistics of resulting arrays are provided in rows 42–43.

4. Results

Seven solutions of optimal coordination with Standard Inverse (SI), reported in the literature over the same 8-bus test case [7] but using different optimization procedures, were taken as examples: LP-SI-[12], NLP [12], NLP-SI-[11], NLP-SI-[13], IGSO-SI-[14], LP-SI-[15] and LP-SI-[16]. NLP means Non-Linear Programming; LP means Linear Programming and IGSO means Improved Group Search Optimization. The input data, from them to our assessment methodology, simply are the results of coordination (the time dial settings $\mathbf{D}$ and pickup settings $\mathbf{P}$). Ref. [16] also reports results of coordination with Very Inverse (VI) and Extremely Inverse (EI) curves, and they are herein called LP-VI-[16] and LP-EI-[16], respectively.

The test-system comprises 8 buses ($b=1, \dots, n_b=8$), 7 transmission lines ($k=1, \dots, n_k=7$) with 2 generation sources ($n_g=2$). Series impedances $z_k \in \mathbf{Z}_K$, shunt admittances $y_k \in \mathbf{Y}_K$ and equivalent impedances $z_g \in \mathbf{Z}_G$ were obtained from [7]. Each line of the system is protected by two DOCP and the corresponding circuit breakers, one at each line end.

A total of $n_r=14$ relays need to be coordinated. The set of relay pairs is $\mathbf{NP} = [3 \ 2 \ 2 \ 3 \ 3 \ 3 \ 4]$. The system has $n_{pr}=20$ primary-backup relay pairs, and the structure of relay-pairs for each line k ($s = 1, \dots, p(k)$) is: Line 1 (2–1, 2–7, 9–10), Line 2 (3–2, 10–11), Line 3 (4–3, 11–12), Line 4 (5–4, 12–13, 12–14), Line 5 (6–5, 6–14, 13–8), Line 6 (1–6, 8–7, 8–9), and Line 7 (14–13, 14–15, 7–9, 7–4). The directional zone (DZ) is set in all relays as $-45^\circ < \angle(V \cdot I_{rkh}^*) < 135^\circ$ with $V=1+0j$ pu for the sake of simplicity. The safety margin is set in $S_M = 300$ ms, which was the value used in six of these seven published examples for their optimal coordination.

4.1. Results considering $n_f=1000$ faults per line

The algorithm shown in Fig. 4 was applied considering $f = 1, \dots, n_f = 1000$ uniformly distributed faults per line, and the 20 relay pairs described in the previous paragraph. That is, the fault location h goes from 0.009% to 99.99% of each line length, and the total number of analyzed relay pairs' behaviors is $N = 20 \cdot 1000 = 20000$.

Results of relay pair classification and the proposed performance indexes are presented in Tables 3 and 4, respectively. The procedure was scripted in Matlab/GNU Octave interpreters using an Apple Mac-BookPro (MacOS Catalina) Intel-i7 2.6 GHz with 16 MB RAM. Subtypes 5b (case 8), 5c (case 9), 6a (case 10), 6b (case 11), and 6e (case 14) do not yield any occurrence in none of the methods evaluated and results are not shown in Table 3. As expected, the sum of all entries from N_1 to N_{15} equals $N = 20000$ in all methods. This means that all the relay pairs for all the analyzed faults have been duly identified and classified.

Results associated with method LP-SI-[15] show a large number of relay pairs classified as Type 3a (case 3), 5a (case 7), 6c (case 12), and 6f (case 15). This result implies no-sensitivity in a vast number of backup relays. In particular, N_{12} yields 6990 occurrences and $N_{\text{nosel}} = 11757$. This imply the worst result from sensitivity standpoint since almost 60% of the analyzed faults cannot be detected by backup relays j in period 2. On the other hand, notice that LP-SI-[12] and the three results from [16] yield the lowest number for $N_{\text{nosel}} = 1161$ with $N_{12} = 438$ occurrences in the case of SI curves. This result is the best solution from sensitivity standpoint.

Method IGSO-[14] shows the largest number of relay pairs with no selectivity ($N_{\text{nosel}} = 6340$). On the other hand, LP-SI-[16] yields the lowest number of no selective relay pairs ($N_{\text{nosel}} = 2062$). Transient configurations due to sequential tripping were only considered in [16], but that coordination did not consider the different fault locations herein assessed. The selectivity index of LP-SI-[12] ($N_{\text{nosel}} = 2392$) is very good, especially because transient configurations due to sequential tripping were not considered in that coordination.

The calculated performance indexes for sensitivity, selectivity and are given in Table 4. Despite all methods claim to be optimal, no one assures complete selectivity and sensitivity.

Algorithm

1: **Inputs:**

$z_k \in \mathbf{Z}_K, y_k \in \mathbf{Y}_K, \forall k \in \mathbf{K}, z_g \in \mathbf{Z}_G, \forall g \in \mathbf{G}, \forall b \in \mathbf{B} \ h \in \mathbf{H}$ [Power system model data]

$\alpha_{r1}, \alpha_{r2} \in \alpha_R, D_r \in \mathbf{D}, P_r \in \mathbf{P}, \forall r \in \mathbf{R}$ [Relay settings data]

$\mathbf{NP} = [p(1), \dots, p(n_k)], S_M$, Main-backup relay-pair table. [Relay-pair data]

2: **for** $f = 1$ to n_f **do** [n_f faults are simulated]

3: $h = f/(1 + n_f)$ [Fault location h is assigned in per unit of total line length]

4: **for** $k = 1$ to n_k **do** [n_k lines are analyzed]

5: $\mathbf{Y}_{bus,f} \leftarrow f(h, \mathbf{Z}_G, \mathbf{Z}_K, \mathbf{Y}_K)$ [Ybus (static) including each location h as an additional bus]

6: **for** $s = 1$ to $p(k)$ **do** [$p(k)$ relay pairs per line k are analyzed and classified]

7: $I_{rkh,s}, \Theta_{rkh,s} \leftarrow f(s, k, h, \mathbf{Y}_{bus,f})$ [Current/angle at primary relays]

8: $\beta_{rkh,s} \leftarrow f(I_{rkh,s}, P_r, \alpha_{r1}, \alpha_{r2})$ [Coefficients β at primary relays]

9: $T_{rkh,s} \leftarrow f(\beta_{rkh,s}, D_r)$ [Operation times at primary relays, Eq. 1]

10: For a relay pair s of line k find the faster primary relay r and tag it as q

11: Reminder relays of pair s are also analyzed and tagged as i, j , and p

12: $\beta_{qkh,s}, \beta_{ikh,s} \leftarrow f(I_{qkh,s}, P_q, I_{ikh,s}, P_i, \alpha_{q1}, \alpha_{q2}, \alpha_{i1}, \alpha_{i2})$ [Coefficients β at primary relays]

13: $\mathbf{Y}_{bus,f} \leftarrow f(h, q, \mathbf{Z}_G, \mathbf{Z}_K, \mathbf{Y}_K)$ [Ybus (transient) when fault contribution from main relay q was cleared]

14: $I'_{ikh,s}, \Theta'_{ikh,s} \leftarrow f(s, i, k, h, \mathbf{Y}_{bus,f})$ [Current/angle at main relays]

15: $I'_{jkh,s}, \Theta'_{jkh,s} \leftarrow f(s, j, k, h, \mathbf{Y}_{bus,f})$ [Current/angle at backup relays]

16: $\beta'_{ikh,s} \leftarrow f(I'_{ikh,s}, P_i, \alpha_{i1}, \alpha_{i2})$ [Transient coefficients β' at main relays]

17: $\beta'_{jkh,s} \leftarrow f(I'_{jkh,s}, P_j, \alpha_{j1}, \alpha_{j2})$ [Transient coefficients β' at backup relays]

18: $\gamma_{qjikh,s} \leftarrow f(\beta_{qkh,s}, \beta_{ikh,s}, \beta'_{ikh,s}, \beta_{jkh,s}, \beta'_{jkh,s})$ [Transient coefficient γ]

19: Relay pair s is classified as subtype from 1 (case $c = 1$) to 6f (case $c = 15$) according to Table 2

20: $N_{c,f,k,s} = 1$ [Only if relay pair s matches one subtype/case c]

21: $N_{f,k,s} = [N_{1,f,k,s}, \dots, N_{c,f,k,s}, \dots, N_{15,f,k,s}]$ [Array with 15 elements]

22: $T_{qkh,s}, T_{pkh,s}, T_{ikh,s}, T_{jkh,s}$ [Eq. 1] [Operation times in Period 1]

23: $T'_{ikh,s}, T'_{jkh,s} \leftarrow f(\beta'_{ikh,s}, \beta'_{jkh,s}), D_i, D_j$ [Transient operation times in Period 2, Eq. 1]

24: $T_{ikh,s}^*, T_{jkh,s}^* \leftarrow f(\gamma_{qjikh,s})$ [Effective operation times at Period 2, Section 3.5]

25: $S_{jikh,s}$ and $S_{pqkh,s}$ [Separation times, Section 3.5]

26: **end for**

27: $N_{c,f,k} = \sum_{s=1}^{p(k)} N_{c,f,k,s} \forall c = 1, \dots, 15$

28: $\mathbf{N}_{f,k} = [N_{1,f,k}, \dots, N_{c,f,k}, \dots, N_{15,f,k}]$ [Array with 15 elements]

29: $\mathbf{S}_{f,k} \leftarrow f(S_{jikh,s}, S_{pqkh,s})$ [Array with $p(k)$ elements]

30: $\mathbf{T}_{f,k} = [T_{ikh,s}, T_{qkh,s}]$ [Array with 2 elements]

31: **end for**

32: $N_{c,k} = \sum_{f=1}^{n_f} N_{c,f,k} \forall c = 1, \dots, 15$

33: $\mathbf{N}_f = [N_{1,1}, \dots, N_{c,k}, \dots, N_{15,n_k}]$ [Array with 15 elements]

34: $\mathbf{S}_f \leftarrow \mathbf{S}_{f,1} \cup \mathbf{S}_{f,2}, \dots, \mathbf{S}_{f,k}, \dots, \mathbf{S}_{f,n_k}$ [Array with $n_{pr}=p(1) + \dots + p(n_k)$ elements]

35: $\mathbf{T}_f \leftarrow \mathbf{T}_{f,1} \cup \mathbf{T}_{f,2}, \dots, \mathbf{T}_{f,k}, \dots, \mathbf{T}_{f,n_k}$ [Array with $2n_k$ elements]

36: **end for**

37: $N_c = \sum_{f=1}^{n_f} N_{c,f} \forall c = 1, \dots, 15$

38: $\mathbf{N} = [N_1, \dots, N_c, \dots, N_{15}]$ [Array with 15 elements]

39: $\mathbf{S} \leftarrow \mathbf{S}_1 \cup \mathbf{S}_2, \dots, \mathbf{S}_f, \dots, \mathbf{S}_{n_f}$ [Array with $n_f n_{pr}$ elements]

40: $\mathbf{T} \leftarrow \mathbf{T}_1 \cup \mathbf{T}_2, \dots, \mathbf{T}_f, \dots, \mathbf{T}_{n_f}$ [Array with $2n_k n_f$ elements]

41: $N_F, N_{NF}, N_{nosen}, N_{nosel} \leftarrow f(\mathbf{N}, S_M)$ [Eq. 2, Eq. 3, and Eq. 4]

42: $[\bar{T}, \sigma_T] \leftarrow f(\mathbf{T})$ [Stats of $\mathbf{T}$]

43: $[\bar{S}, S_{min}, S_{max}] \leftarrow f(\mathbf{S}, S_M)$ [Stats of $\mathbf{S}$, histograms]

44: **Outputs:**

sensitivity index $\leftarrow f(N_{nosen}, N_F)$ [Eq. 10]; selectivity index $\leftarrow f(N_{nosel})$ [Eq. 11]

primary operation time $\leftarrow (\bar{T})$ [Eq. 12]; speed index $\leftarrow f(\bar{T})$ [Eq. 13]

Fig. 4. Algorithm to compute the performance indexes of coordination of DOCP.**Table 3**

Relay-pair classification results — total occurrences per subtype/case.

Methods under evaluation	Relay pair subtype/case										N_{nosel}	N_{nosen}
	1	2	3a	3b	3c	4	5a	6c	6d	6f		
	N_1	N_2	N_3	N_4	N_5	N_6	N_7	N_{12}	N_{13}	N_{15}		
LP-SI-[12]	8554	8194	721	1548	531	0	0	440	12	0	2392	1161
NLP-SI-[11]	7713	6952	2259	985	1070	0	0	985	36	0	2945	3244
NLP-SI-[12]	7806	7029	2139	1022	1017	0	0	935	52	0	2860	3074
NLP-SI-[13]	7067	6298	2953	656	1435	0	0	1591	0	0	4550	4544
IGSO-SI-[14]	6977	5193	886	370	1243	658	0	3103	432	1138	6340	5127
LP-SI-[15]	2796	2067	1284	248	752	2289	2096	6990	91	1387	2705	11757
LP-SI-[16]	8564	8184	723	1548	529	0	0	438	14	0	2062	1161
LP-VI-[16]	8576	8172	711	1548	541	0	0	450	2	0	2994	1161
LP-EI-[16]	8689	8059	749	1548	503	0	0	412	40	0	3885	1161

Table 4
Calculated performance indexes and other relevant indicators.

	Methods under evaluation								
	LP-SI-[12]	NLP-SI-[11]	NLP-SI-[12]	NLP-SI-[13]	IGSO-SI-[14]	LP-SI-[15]	LP-SI-[16]	LP-VI-[16]	LP-EI-[16]
N	20000	20000	20000	20000	20000	20000	20000	20000	20000
N_F	19548	18979	19013	18409	15327	11532	19548	19548	19548
N_{NF}	452	1021	987	1591	4673	8468	452	452	452
Sensitivity (%)	94.2	83.8	84.6	77.3	74.4	41.2	94.2	94.2	94.2
Selectivity (%)	87.8	84.5	85.0	76.5	58.6	75.2	89.5	84.7	80.1
$\bar{v}$ (1/s)	1.154	1.960	1.943	2.432	3.488	1.387	1.089	2.314	4.256
$\bar{T}$ (ms)	885.0	538.6	540.6	457.1	318.4	1014.3	938.0	478.5	335.8
σ_T (ms)	120.1	118.1	113.9	143.8	97.2	532.5	125.1	139.4	181.2
$\bar{S}$ (ms)	962	860	857	749	2505	6065	1007	1320.1	3099
S_{min} (ms)	233.9	211.3	248.2	168.0	-100.2	-530.6	270.6	276.5	252.4
S_{max} (s)	2654	719	642	297	8225	3892	2756	3935.6	10099
CPU time (s)	34.2	33.2	33.9	33.5	33.1	33.0	33.3	33.4	36.1

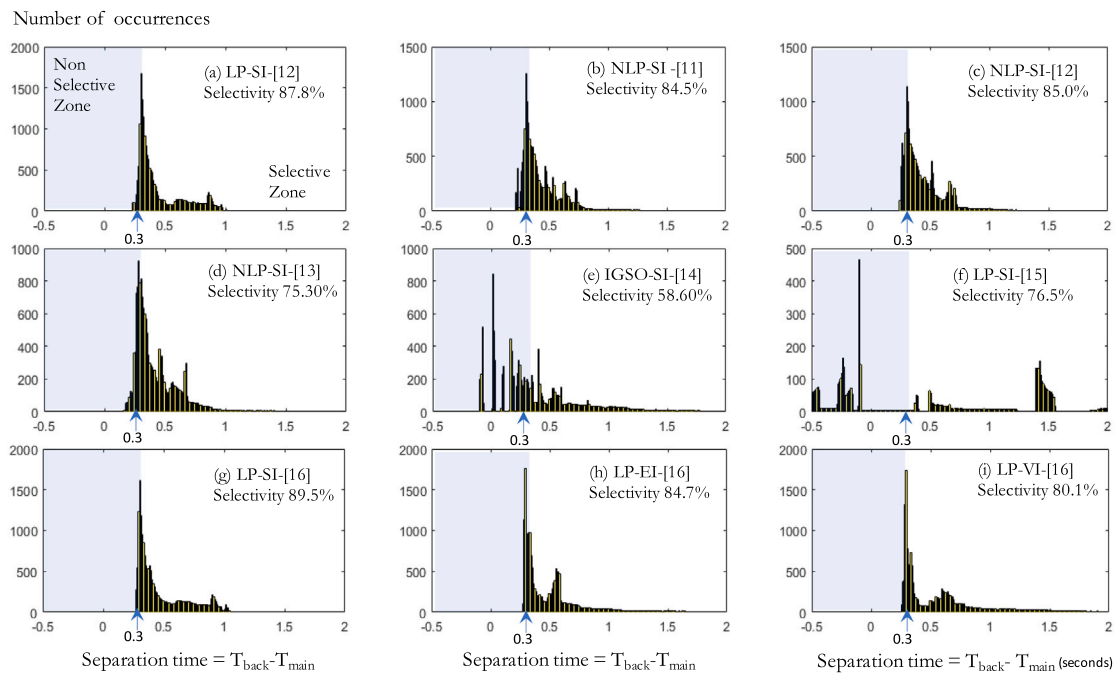


Fig. 5. Histograms for calculated separation times per method evaluated.

It is worth noting that LP methods only seek maximum speed subject to selectivity constraints since these methods assume that the sensitivity problem has been previously solved. Solutions from [12,16] have maximum possible sensitivity, considering the constraints about load currents, but the pickup currents of solution from [15] seem that were selected without considering that sensitivity is a desirable protection feature. On the other hand, NLP methods and other methods maximize speed subject to both selectivity and sensitivity constraints.

The fastest solution (Table 4) in methods that use SI curves corresponds to IGSO-SI-[14] with an average primary relay speed of $\bar{v} = 3.488 \text{ s}^{-1}$ and an average primary relay operation time $\bar{T} = 318 \text{ ms}$. This solution is around three times faster than solutions LP-SI-[12] and LP-SI-[16], but results with settings of IGSO-SI-[14] have the worst selectivity (58.6%). Result with settings of LP-SI-[15] has the worst sensitivity index (41.2%) and LP-SI-[16] the worst speed ($\bar{v} = 1.089 \text{ 1/s}$).

Other relevant statistics are shown in rows 7–10 of Table 4. Standard deviations σ_T of primary operation time array (T) are around 100–140 ms. Solution reported by [15] is out of trend ($\sigma_T = 532 \text{ ms}$, with $\bar{S} = 6000 \text{ ms}$) and severe miscoordination conditions ($S_{min} = -530 \text{ ms}$). These results imply average separation times of 6 s and, in some cases backup relays operating first since the separation time is

negative. Histograms of Fig. 5 are useful to show this type of situations (shaded areas correspond to occurrences where separation times were lower than the safety margin, and negative values correspond to backup relays faster than main relays). Histograms only show occurrences between -0.5 and 2 s to facilitate the readability of graphs.

Notice that the fastest solution IGSO-SI-[14] also shows relevant miscoordination in Fig. 5(e). Slowest methods, such as LP-SI-[12] and LP-SI-[16], reveal the best selectivity indexes (87.8 and 89.5%) and excellent values of S_{min} (233.9 and 270.6 ms). These values of S_{min} are very close to the prescribed coordination time interval (300 ms) in the optimization problems taken as examples. That is, the selectivity is very probable for values of S_{min} greater than 250 ms, even if it is not 100% guaranteed (since S_{min} is not greater than 300 ms). From this perspective, the results of LP-EI-[16], LP-SI-[16], LP-VI-[16], LP-SI-[12] and NLP-SI-[12] are excellent, the results of NLP-SI-[11] and NLP-SI-[13] are not very bad, and the results of IGSO-SI-[14] and LP-SI-[15] are very bad. The histograms of Fig. 5 are also useful to illustrate these facts. S_M is 300 ms in the optimization problems taken as examples (except for LP-SI-[15], that used $S_M = 400 \text{ ms}$ instead), but the selectivity constraints were not formulated for all the possible fault locations.

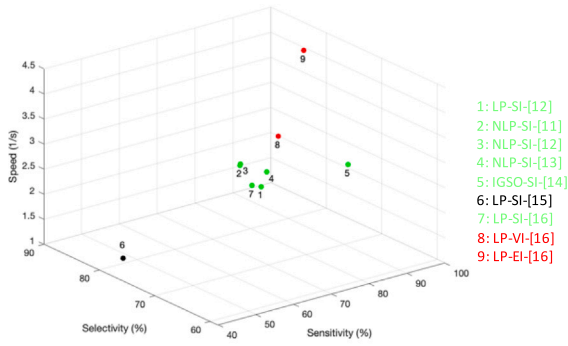


Fig. 6. Resulting trade-offs among selectivity, sensitivity, and speed performance indexes — 3D view.

The indexes for selectivity, sensitivity, and speed of analyzed solutions are herein depicted using three axes to facilitate the comparison among solutions. The indexes are depicted in Fig. 6 as a 3D graph, and in Fig. 7 as the 2D orthographic projections to illustrate the trade-offs between speed/selectivity, speed/sensitivity, and selectivity/sensitivity. Solutions that use the same relay curve (Standard Inverse SI) are depicted as a green bullet (except LP-SI-[15], denoted as a black bullet because it is clearly out of trend).

The green bullets in Figs. 7(a) and (b) tend to follow a decreasing curve, which means that speed can only be improved by decreasing sensitivity and/or selectivity. This behavior is similar to the classical Pareto frontiers of multi-objective optimization. That is, the fastest solution with green bullets (IGSO-SI-[14]) is the worst for sensitivity and selectivity, the slowest solutions (LP-SI-[12] and LP-SI-[16]) are the best for sensitivity and selectivity, and there are other optimal solutions between those extreme cases. In case of sensitivity/selectivity (Fig. 7(c)), the green bullets tend to follow an increasing curve, which means that an increase of sensitivity tends to improve also the selectivity. The black bullet is obviously out of these trends, because the pickup currents of LP-SI-[15] seem that were selected without considering that sensitivity is a desirable protection feature.

Applying multi-objective optimization terminology, the solutions denoted with green bullets can be regarded as efficient (like Pareto non dominated solutions) since some improvement in one objective (speed) implies a detriment, in at least one of the other two objectives (selectivity and sensitivity).

Results depicted in Fig. 7 are good enough to illustrate the benefits of proposed methodology. However, we must stress that any solution specified by any optimal coordination procedure for DOCP systems, present and future, can be benchmarked using the proposed methodology.

It is worth analyzing solutions considering different relay curves. Optimal results with Very Inverse and Extremely Inverse curves (LP-VI-[16] and LP-EI-[16], respectively) are depicted in Fig. 7 as a red bullet. These results show faster solutions, with some deterioration of the selectivity index. The sensitivity index of LP-SI-[16], LP-VI-[16] and LP-EI-[16] is the same since the pickup settings remain unchanged. Fig. 7(a) shows that LP-EI-[16] is faster than LP-VI-[16] but with lower selectivity. That is, the application of the proposed indexes is useful to evaluate and benchmark a set of existing solutions regardless of using or not the same curve types in them. Therefore, we consider the research question stated in the introduction of this paper has been answered.

4.2. Sensitivity analysis

Previously 1000 faults per line were considered in order to obtain certainty about the validity of the results.

However, a lower n_f could be also good enough to obtain reliable results and this point could be important in case of large-scale grids,

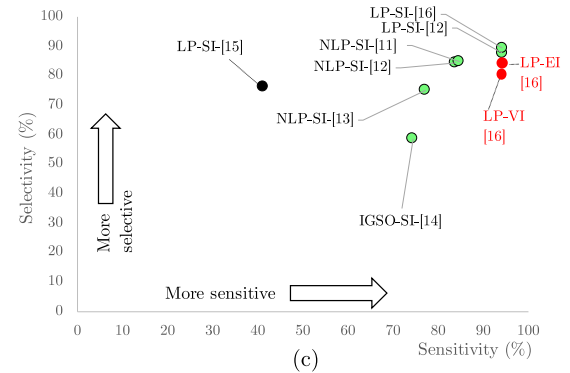
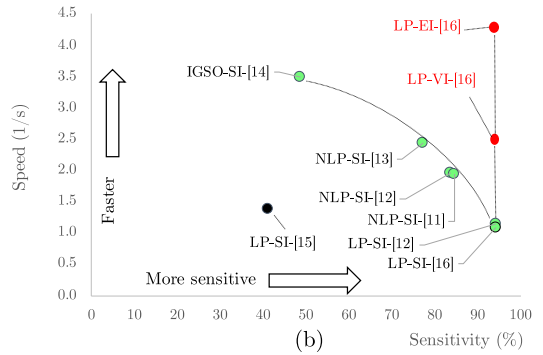
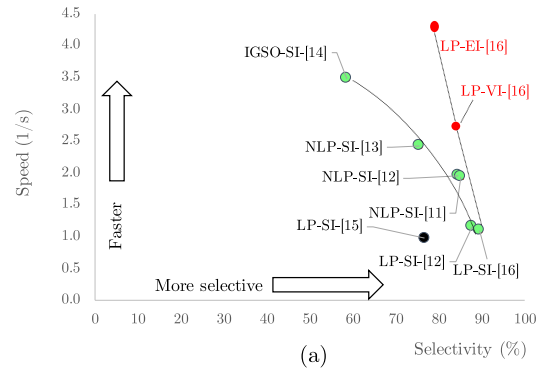


Fig. 7. Resulting trade-offs among selectivity, sensitivity, and speed performance indexes — 2D view.

where the computational burden of using $n_f = 1000$ could be prohibitive (even if this analysis is performed off-line, as usual). The results for a range of n_f from 1 up to 1000 were computed and are shown in Fig. 8. In all the analyzed solutions, the values of the indexes tend to stabilize to the accurate solution from $n_f = 20$ ($N = 400$ relay pairs).

The results of Fig. 8 were obtained by using the expression $h = f/(1+n_f)$ (line 3 of algorithm shown in Fig. 4) as the way of distributing the fault locations. For very low values of n_f , is probably more efficient to include always faults near to both line-ends. An analysis about the optimal way of distributing the fault locations, in order to maximize the accuracy of results with minimum number of fault locations (and considering different power system sizes), should be numerically analyzed in future research about this topic.

4.3. Results considering pre-fault conditions

Previous results (with $n_f = 1000$) were obtained without considering pre-fault conditions to compute fault currents (assumption 3 of Section 3.1). A summary of results considering pre-fault conditions to

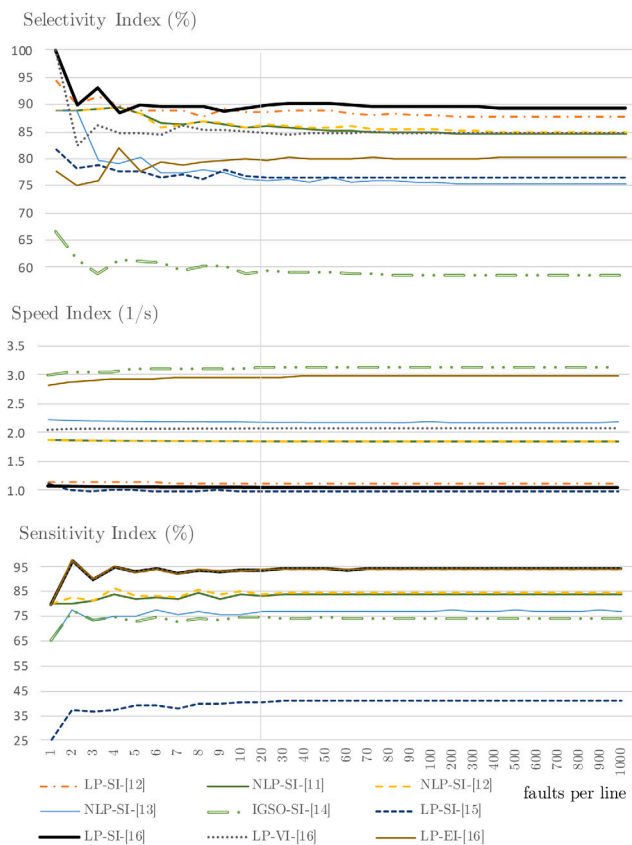


Fig. 8. Effect of varying the number of faults per line on the results of proposed performance indexes.

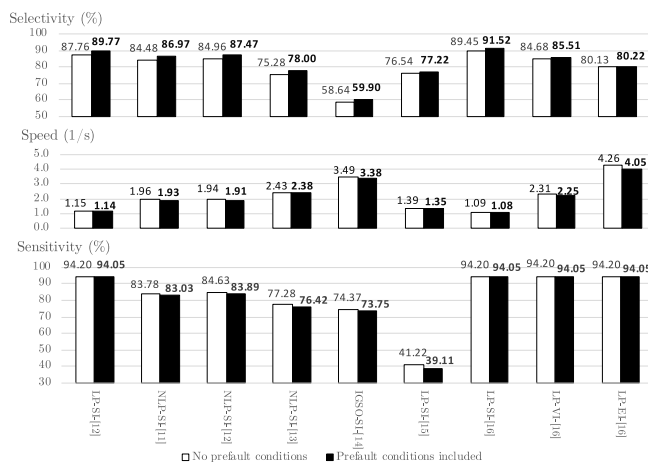


Fig. 9. Performance indexes considering pre-fault conditions.

compute fault currents is shown in Fig. 9 and compared with results of Section 4.1. The inclusion of pre-fault conditions led to lower fault current magnitudes and, consequently, lower sensitivity indexes and slower speed indexes. On the other hand, an increase of the selectivity index is observed in all the analyzed cases. However, the differences between results with and without considering pre-fault conditions to compute fault currents are relatively small for the analyzed cases.

5. Conclusions

There was no published method for an exhaustive evaluation of quality of solutions to the problem of coordination of Directional

Over-Current Protections (DOCP). This paper fulfills this research gap, introducing a novel method which computes selectivity, sensitivity, and speed performance indexes for a wide range of possible fault locations.

The proposed performance indexes are determined as result of a meticulous process of identification and classification of the behavior of relay pairs for each simulated fault, as well as the detailed assessment of the selectivity and the protection speed. Furthermore, the developed method offers an adequate comparison framework by identifying the conflicts among coordination solutions from a multiple-objective optimization perspective.

Seven different optimal solutions for the same problem of DOCP coordination, reported in the literature using the same curve type (IEC standard inverse), were analyzed to illustrate the proposed methodology. In this case, the solutions with higher speeds imply a deterioration of sensitivity and/or selectivity, and this fact implies a behavior similar to a Pareto frontier. Only one of the analyzed optimal solutions does not follow this behavior, and the reason behind this fact is that the pickup currents in this case seem that were selected without considering that sensitivity is a desirable protection feature. No reported solution was selective for 100% of analyzed faults; the reason behind this fact is that the analyzed optimal solutions did not include transient configurations and a wide range of fault locations in their problem formulation.

The proposed method is useful to compare coordination solutions with different curve types. In fact, reported solutions with IEC very inverse and IEC extremely inverse for the same problem were evaluated and the obtained speed, sensitivity and selectivity indexes were compared with those of the seven optimal solutions with IEC standard inverse curve. In the analyzed optimal solutions, all the system relays have the same curve type. However, the proposed method can be also applied in cases of a different curve type for each relay location.

One of the main benefits of the proposed methodology, based on specific numeric indexes, is the objective comparison of different existing solutions for the same (coordination) problem. That is, the outcomes of the methodology are independent of the analyst's subjective criteria since all the solutions under evaluation are analyzed under the same assumptions. Other performance indexes can be defined for selectivity, sensitivity and speed, and the proposed exhaustive assessment method is still conceptually valid. The proposed method also offers an important contribution for the possible future development of multi-objective optimization applied to DOCP coordination since the tool for the proper assessment of solutions has been herein shown (and new problem formulations can be easily based on this).

CRediT authorship contribution statement

P.M. De Oliveira-De Jesus: Data curation, Conceptualization, Methodology, Software, Writing – original draft. **E. Sorrentino:** Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix. Nomenclature

<i>Subindexes</i>	
b	Denotes a transmission system bus
c	Denotes a case number associated with a subtype
f	Denotes a fault number
g	Denotes a generation bus
h	Denotes a fault location
i	Denotes a primary relay that trips after relay q
j	Denotes a backup relay of a primary relay i
k	Denotes a transmission line
q	Denotes a primary relay that trips first than relay i
p	Denotes a backup relay of a primary relay q
s	Denotes a relay-pair
<i>Data sets</i>	
α_R	Set of relay parameters
β'	Set of linear coefficient (transient)
β	Set of linear coefficient (static)
Θ	Set of static fault angles (deg)
Θ'	Set of transient fault angles (deg)
φ	Set of analyzed faults with no-selective relay pairs
B	Set of transmission buses
D	Set of optimal time dial settings
F	Set of analyzed faults with feasible relay pairs
G	Set of generation buses
H	Set of fault locations
I	Set of static fault current magnitudes (A)
I'	Set of transient fault current magnitudes (A)
K	Set of transmission lines
N	Set of analyzed relay pairs for the simulated faults
NP	Set of relay pairs
NF	Set of analyzed relay pairs' behaviors without computable separation time
NL	Set of analyzed relay pairs' behaviors without sensitivity in Period 1 or Period 2
P	Set of pick-up current settings (A)
T	Set of primary relay operation times (s)
R	Set of relays
S	Set of separation times between backup and primary relays (s)
Y_K	Set of shunt line admittances (Ω^{-1} , pu)
Z_K	Set of series line impedances (Ω , pu)
<i>Matrices</i>	
Y_{bus}	Static system admittance matrix (Ω^{-1} , pu)
Y'_{bus}	Transient system admittance matrix (Ω^{-1} , pu)
<i>Variables</i>	
α	Relay curve parameter
β'	Linear coefficient (transient) (s)
β	Linear coefficient (static) (s)
γ	Linear transient factor
σ	Standard deviation
θ	Static fault angles (deg)
θ'	Transient fault angles (deg)
$\bar{v}$	Average speed index (1/s)
I	Static fault current magnitude (A)
I'	Transient fault current magnitude (A)

n_b	Number of transmission buses
N_F	Analyzed relay pairs' behaviors with calculated separation time
N_{nosen}	Analyzed faults with no-sensitive backup relays
N_{nosel}	Analyzed faults with no-selective relay pairs
N_{NF}	Analyzed relay pairs' behaviors without computable separation time
N	Total number of analyzed relay pairs' behaviors
n_{pr}	Number of total primary-backup relay pairs
n_f	Number of faults uniformly distributed
n_k	Number of transmission lines
n_g	Number of generators
n_r	Number of relays
D_r	Time dial setting of relay r
P_r	Pick-up current setting of relay r (A)
$p(k)$	number of relay pairs in line k
S	Separation time (s)
S_M	Safety margin (s)
S_{min}	Minimum separation time (s)
S_{max}	Maximum separation time (s)
T	Static operation time (s)
T'	Transient operation time (s)
T^*	Effective transient operation time (s)
$\bar{T}$	Average primary relay operation time (s)
z_k	π model, series impedance (Ω , pu)
y_k	π model, shunt admittance (Ω^{-1} , pu)

References

- [1] A.J. Urdaneta, R. Nadira, L.G. Perez Jimenez, Optimal coordination of directional overcurrent relays in interconnected power systems, *IEEE Trans. Power Deliv.* 3 (3) (1988) 903–911.
- [2] M.H. Hussain, S.R.A. Rahim, I. Musirin, Optimal overcurrent relay coordination: a review, *Procedia Eng.* 53 (1) (2013) 332–336.
- [3] A.R. Al-Roomi, Optimal Coordination of Power Protective Devices with Illustrative Examples, John Wiley & Sons, 2021.
- [4] S. Raza, T. Mahmood, S. Bukhari, Optimum overcurrent relay coordination: A review, *Nucleus* 51 (1) (2014) 37–49.
- [5] D. Birla, R.P. Maheshwari, H.O. Gupta, Time-overcurrent relay coordination: A review, *Int. J. Emerg. Elect. Power Syst.* 2 (2) (2005).
- [6] S.D. Godwal, K.S. Pandya, V.N. Rajput, S.C. Vora, A review on approaches employed for solving directional overcurrent relays' coordination problem, in: *Advances in Electric Power and Energy Infrastructure*, Springer, 2020, pp. 35–51.
- [7] A. Braga, J. Saraiva, Coordination of overcurrent directional relays in meshed networks using the simplex method, in: *Proceedings of 8th MELECON 96*, vol. 3, IEEE, 1996, pp. 1535–1538.
- [8] A. Darabi, M. Bagheri, G. Gharehpetian, Dual feasible direction-finding nonlinear programming combined with metaheuristic approaches for exact overcurrent relay coordination, *Int. J. Electr. Power Energy Syst.* 114 (2020) 105420.
- [9] E. Sorrentino, J.V. Rodríguez, Effects of the curve type of overcurrent functions and the location of analyzed faults on the optimal coordination of directional overcurrent protections, *Comput. Electr. Eng.* 88 (2020) 106864.
- [10] D.F.C. Rodriguez, J.D.P. Osorio, G. Ramos, Virtual relay design for feeder protection testing with online simulation, *IEEE Trans. Ind. Appl.* 54 (1) (2017) 143–149.
- [11] D. Birla, R.P. Maheshwari, H. Gupta, A new nonlinear directional overcurrent relay coordination technique, and banes and boons of near-end faults based approach, *IEEE Trans. Power Deliv.* 21 (3) (2006) 1176–1182.
- [12] M. Ezzeddine, R. Kaczmarek, M. Iftikhar, Coordination of directional overcurrent relays using a novel method to select their settings, *IET Gener. Transm. Distrib.* 5 (7) (2011) 743–750.
- [13] A. Mahari, H. Seyedi, An analytic approach for optimal coordination of overcurrent relays, *IET Gener. Transm. Distrib.* 7 (7) (2013) 674–680.
- [14] M. Alipour, S. Teimourzadeh, H. Seyedi, Improved group search optimization algorithm for coordination of directional overcurrent relays, *Swarm Evol. Comput.* 23 (2015) 40–49.
- [15] M. Meskin, A. Domijan, I. Grinberg, Optimal co-ordination of overcurrent relays in the interconnected power systems using break points, *Electr. Power Syst. Res.* 127 (2015) 53–63.
- [16] E. Sorrentino, J.V. Rodríguez, A novel and simpler way to include transient configurations in optimal coordination of directional overcurrent protections, *Electr. Power Syst. Res.* 180 (2020) 106127.

- [17] C.A.C. Coello, G.B. Lamont, D.A.V. Veldhuizen, Multi-criteria decision making, in: *Evolutionary Algorithms for Solving Multi-Objective Problems: Second Edition*, Springer, 2007, pp. 515–545.
- [18] M. Singh, B.K. Panigrahi, A.R. Abhyankar, S. Das, Optimal coordination of directional over-current relays using informative differential evolution algorithm, *J. Comput. Sci.* 5 (2) (2014) 269–276.
- [19] T.A. Abd Almuhsen, A.J. Sultan, Using of genetic algorithm to obtain proper coordination of directional overcurrent relays, *IOP Conf. Ser.: Mater. Sci. Eng.* 881 (1) (2020) 012131.
- [20] C.-R. Chen, C.H. Lee, C.-J. Chang, Overcurrent relay coordination optimization with partial differentiation approach for the validation of coordination violation, *Electr. Power Compon. Syst.* 39 (10) (2011) 933–947.
- [21] P. Kumar, V. Kumar, R. Pratap, FPGA based design and HIL verification of a soft-core directional-OCR, *Electr. Power Compon. Syst.* 47 (14–15) (2019) 1423–1436.
- [22] M. Latroch, M. Khat, D. Rahiel, An IDMT overcurrent protective relay based on ADALINE, *Eur. J. Elect. Eng.* 21 (6) (2019) 515–522.
- [23] A. Mahboubkhah, V. Talavat, M. Beiraghi, Considering transient state in interconnected networks during fault for coordination of directional overcurrent relays, *Electr. Power Syst. Res.* 186 (2020) 106413.
- [24] V.A. Papaspiliotopoulos, G.N. Korres, V.A. Kleftakis, N.D. Hatzargyriou, Hardware-in-the-loop design and optimal setting of adaptive protection schemes for distribution systems with distributed generation, *IEEE Trans. Power Deliv.* 32 (1) (2015) 393–400.
- [25] J.J. Grainger, W.D. Stevenson, W.D. Stevenson, et al., *Power System Analysis*, 2003.
- [26] A.J. Urdaneta, L.G. Pérez, H. Restrepo, Optimal coordination of directional overcurrent relays considering dynamic changes in the network topology, *IEEE Trans. Power Deliv.* 12 (4) (1997) 1458–1464.